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Blessed G/Michael Catholic Secondary School
2nd Semester Chemistry test for Grade 12th

Mar, 2018 E.C Time Allowed - 30 min

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I. Choose the best answer from the given alternatives.

C 1. The degree of polymerization (DP) is defined as:

A .Number of polymer chains in a sample C. Number of monomer units in a polymer chain
B. Molecular weight of polymer D. Number of branches in polymer

B 2. In addition polymerization, the polymer forms via:

A. Condensation reaction with elimination of water C. Formation of ionic salts
B. Repeated addition of monomers without by-products D .Esterification reaction

C 3. Which of the following is a condensation polymer?

A .Polyethylene B .Polyvinyl chloride C. Nylon-6,6 D. Polystyrene

B 4. The three main steps of addition polymerization are:

A. Activation, propagation, termination C. Condensation, chain growth, branching
B. Initiation, propagation, termination D. Initiation, branching, crosslinking

B 5. Polypropylene is classified as:

A .Thermosetting polymer B. Thermoplastic polymer C. Elastome D. Biopolymer

B 6. A polymer that softens on heating and hardens on cooling is:

A .Thermosetting B. Thermoplastic C. Elastomer D .Dendrimer

A 7. The monomer of polystyrene is:

A. Vinyl benzene B . Ethylene C. Propylene, D .Vinyl chloride

A 8. Nylon-6,6 is synthesized from:

A. Hexamethylenediamine and adipic acid C. Ethylene glycol and terephthalic acid
B. Caprolactam only D . Lactic acid

B 9. Which property is typical of thermosetting polymers?

A. Softens on heating B. Insoluble and infusible C. Highly flexible D. Linear structure

A 10. The molecular weight of a polymer is:

A. molecular weight of monomers × degree of polymerization C. Always constant
B . Equal to DP only D. Equal to monomer mass only

A 11. Polyesters are formed by:

A .Esterification of diacid and diol C. Radical polymerization
B. Addition of vinyl monomers D. Ionic polymerization

A 12. The initiator in radical polymerization serves to:

A. Start the reaction by forming radicals C. Crosslink monomers
B . Terminate the polymer chain D. Convert thermoplastics into thermosets

B 13. Teflon (PTFE) is made from:

A. Vinyl chloride B. Tetrafluoroethylene C. Styrene D. Ethylene oxide

A 14. Which polymer contains amide linkages (-CONH-)?

A. Nylon B. Polyester C. Polyurethane D. Polystyrene

A 15.. Which polymer is used in making bottles and food packaging?

A.PET B. Bakelite C. Polyurethane D .Nylon-66

D 16. Natural rubber is:

A. neoprene B. Polyethylene C. Polystyrene D. Polyisoprene

A 17.. Vulcanization of rubber involves:

A. Crosslinking with sulfur C. Condensation polymerization

B. Addition polymerizatio D. Hydrogenation

B 18. Monomers in condensation polymerization typically have:

A. One functional group C. Only vinyl groups

B. Two or more reactive functional group D. Aromatic rings only

D 19. The "propagation" step in radical polymerization is:

A . Formation of radicals C. Breaking of polymer chains

B. crosslinking D. Growth of polymer chain by successive addition of monomers

A 20. Which polymer is highly resistant to chemicals and heat?

A. PTFE (Teflon) B.LDPE C. Polystyrene D. Nylon-66

B 21. Thermosetting polymers are:

A. Linear B. Crosslinked C. Branched D. Always elastomers

B 22. The term "copolymer" refers to:

A. Polymer from single monomer C .Biopolymer

B. Polymer from two or more different monomers D. Elastomer

B 23. The polymerization of styrene ($C_6H_5–CH=CH_2$) to form polystyrene ($–CH_2–CH(C_6H_5)–)_n$ occurs via addition polymerization. Which of the following statements is correct about this reaction?

A. Styrene reacts with another styrene molecule by elimination of water to form polystyrene.

B. Polystyrene forms by repeated addition of styrene monomers through the breaking of the double bond without by-products.

C. Polystyrene is formed by condensation between styrene and benzene with elimination of HCl.

D. Styrene polymerizes via ring-opening polymerization of its aromatic ring

C 24. What is the monomer of polypropylene?

A . $CH_2=CH_2$ B. $CH_2=CHCH_3$ C. $CH_3–CH_3$ D. $CH_2=CHCl$

25 ———.used as a substitute for glass?

A. Polyethylene B. Bakelite C. Perspex (PMMA) D. Nylon